

NEXUS

The multi-model inference engine for hardware you already own

Free. Local. Portable. Nexus brings real large-language-model performance to a single modest graphics card, a drawer of old ones, an old laptop, or a CPU alone — and links spare computers together when one isn't enough. No subscription, no cloud, nothing leaves your machine.

Multi-model | GPU-accelerated prefill | Persistent KV memory | Every graphics card, old ones too | Multi-computer distributed inference

LOCAL AI · DEMOCRATIZED

◆ WHAT IS NEXUS

One engine, built for the hardware you have

Most local AI tools assume you own a graphics card built for the job. Nexus assumes you don't — and is engineered from the ground up to get real results out of whatever you've actually got.

Run several models at once

Nexus is a **multi-model inference engine** — load more than one model into completely independent slots and run them side by side, each on its own port, each with its own graphics-card placement, sampling settings, and conversation memory. Compare two models head to head, or just keep a small fast model and a larger careful one both warm and ready at the same time.

Prefill acceleration

Reading a long prompt — "prefill" — is normally the slowest, most hardware-hungry part of talking to a large model, and the part that suffers most on modest hardware. Nexus's prefill acceleration streams the model's layers through a graphics card *one at a time* while the prompt is read, then hands off to the CPU (or a GPU-held copy) to generate the reply. The card never needs to hold the whole model — it only ever needs room for a single layer at once.

> The result: GPU-class prompt-reading speed from a card nowhere near big enough to hold the model itself.

◆ HOW IT STAYS FAST

Never pay the same compute cost twice

Two more ideas sit underneath everything Nexus does — one about memory, one about sharing hardware you already own.

Custom KV retention

Most local chat tools re-read the entire conversation from the first token, every single turn. Nexus doesn't. Its KV cache — the model's own working memory of everything it has already read — is retained and reusable. Hand it a huge document, wait once for it to actually be read, and every short follow-up question after that skips the re-read entirely. Save that state to disk, and reloading it drops you straight back into the conversation where you left off — no re-processing, no waiting, just the answer.

Two computers, one model

Short on memory for a bigger model, but there's a second computer sitting around doing nothing? Link them together. Nexus splits a single model's layers across two or more machines over the network, turning a pair of ordinary computers into something neither one could run alone.

The benefit isn't only extra capacity. If the main computer has even a single small graphics card — as little as 1 GB — it can still accelerate prefill for the *whole* setup, prompt-reading speed included, even while the second, donor computer has no graphics card at all. The slower machine only has to carry its share of actually generating the reply, where it counts least.

◆ BUILT FOR REAL WORK

Everything you'd want from it

Fast is the start. Nexus is built to live on: long projects, piles of documents, every card in the case, and other devices on your network.

Conversation memory you control

Forget any message and everything after it shifts back exactly into place. Archive a finished piece of work — a manual, a code revision — and restore it later into any chat, with no re-reading. When a conversation fills up, the oldest messages give way and the chat keeps going.

Drop in your documents

PDFs, Word files, spreadsheets, slides and e-books are read as text right in the chat, with OCR for scanned pages. Drop a whole folder of specs and notes and the model gets all of it, each file under its path. Offline, on every system.

Every graphics card you have

Mix any brands and ages. Each card holds as many layers as really fit, and every speed option is measured on your hardware rather than guessed. Speculative decoding is lossless: the exact same text, sooner.

Safe on your network, easy to build on

Chat from any device at home; websites on the internet can't touch your models. An optional API key, a full streaming HTTP API, and a C++ engine you can link directly.

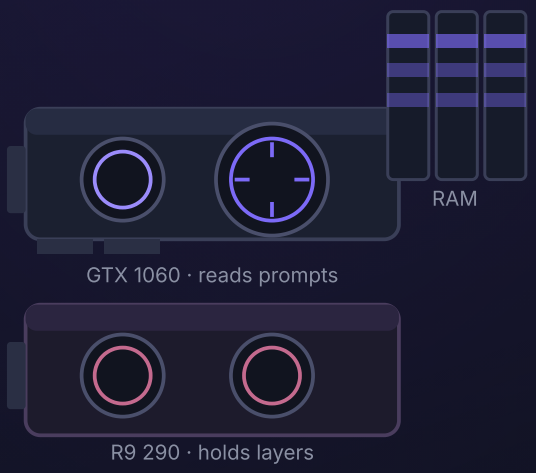
The Code Hardener

Point local models at a whole codebase and walk away: security audits, notes sweeps or your own prompt chains, chunk by chunk, with a final report from your strongest model.

◆ REAL HARDWARE, REAL RESULTS

This isn't a benchmark machine

None of this is theoretical — it's what Nexus is built and tested on. A graphics card from 2016, a $20 used card beside it, off-the-shelf memory, and a 27-billion-parameter model that has no business running this well on hardware like this.



THE TEST RIG

MEMORY	25 GB DDR4 @ 2200 MHz
PROMPT CARD	3 GB — GeForce GTX 1060
HELPER CARD	4 GB — Radeon R9 290 ($20)
MODEL	Qwen 3.6 27B, 4-bit
READING	42–50 tok/sec
WRITING	over 1 tok/sec

An 11,400-token prompt read in about four and a half minutes, 60,000-token prompts still above 30 tokens per second, and the desktop stays usable the whole time — on a 3 GB card from 2016, a used card that cost less than lunch, and 25 GB of ordinary memory. No data-center GPU, no multi-thousand-dollar rig. This is the entire point of Nexus: the hardware most people already own, put to real use.

3+4 GB VRAM
TWO OLD GRAPHICS CARDS

27B params
MODEL SIZE (4-BIT)

~45 tok/s
PROMPT READING

Get started

Nexus is free and runs entirely on your own machine. Pick a model, point Nexus at it, and you're generating — no account, no cloud round-trip, nothing phoned home.

The Nexus Guide
Everything in one place: setups, every setting, flag and endpoint.

Launcher Guide
First run through picking a model, GPU placement, and starting your first chat.

Quant Builder Guide
Converting and packaging your own models into Nexus's native format.

Graphics Cards & Speed
Several cards, 8-bit math, MTP, and getting the most from your slots.

Conversation Memory
Limits, auto-trim, and forgetting, archiving and restoring.

Engine / API Guide
Running headless, the full HTTP API, and linking the engine directly.

Released under AGPLv3 · github.com/sjnettifee/nexus